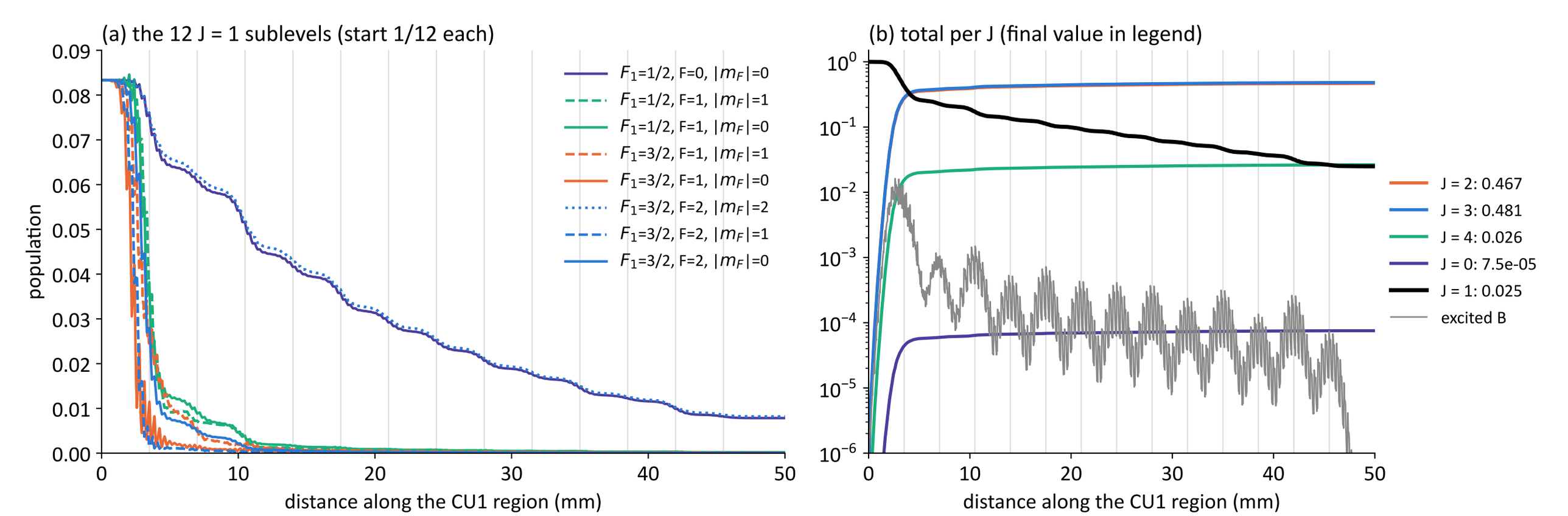

(a) the 12 J = 1 sublevels (start 1/12 each)
population
distance along the CU1 region (mm)
F_1=1/2, F=0, $|m_F|$=0
F_1=1/2, F=1, $|m_F|$=1
F_1=1/2, F=1, $|m_F|$=0
F_1=3/2, F=1, $|m_F|$=1
F_1=3/2, F=1, $|m_F|$=0
F_1=3/2, F=2, $|m_F|$=2
F_1=3/2, F=2, $|m_F|$=1
F_1=3/2, F=2, $|m_F|$=0
(b) total per J (final value in legend)
distance along the CU1 region (mm)
J = 2: 0.467
J = 3: 0.481
J = 4: 0.026
J = 0: 7.5e-05
J = 1: 0.025
excited B